9 The Circular Restricted Three–Body Problem

9.1 Introduction

In Chapters 4 and 5 we have seen that the preliminary design of orbital transfers and interplanetary trajectories is usually established on the Two–Body Problem, that is, on the assumption that the spacecraft is attracted gravitationally only by one central body at a time. In a second step, the motion of the probe is refined by adding further perturbations, such as the ones due to the presence of other bodies. Nowadays, this classical approach can be supported by innovative dynamical techniques, in order to meet the increasing challenges faced by space explorations missions. In particular, a given trajectory can be designed on the basis of both impulsive burns and low-thrust arcs and, at a some point of the mission, the Keplerian orbits can be replaced by other kinds of solutions, whose existence can be proved by assuming that two bodies attract gravitationally the probe at the same time with a comparable effect. The latter assumptions correspond to the *Circular Restricted Three–Body Problem* (CR3BP) dynamical model, which recently has been applied to various types of celestial mechanics problems, related both to natural and artificial bodies. In this Chapter, we will present it, being the aim to enlarge our knowledge regarding the possible solutions which are available to the orbit designer. In many cases, the classical approach and the CR3BP are complementary and therefore we should not look at them as in contrast.

Unlike the Two–Body Problem, the CR3BP, as well as the general Three–Body Problem, does not represent an integrable system, that is, it is not possible to find a general solution to the equations of motion. However, in a proper reference frame we are able to introduce some mathematical objects which help at simplifying the analysis. These are in particular the Jacobi constant, the equilibrium points and the corresponding periodic or quasi-periodic orbits and hyperbolic invariant manifolds.

Here we will just rough out periodic orbits and invariant manifolds, but it is important to understand that all the successful applications of the CR3BP are based on them. To mention some, the dynamical evolution of some comets due to the combined gravitational attractions by the Sun and Jupiter and the motion of asteroids in the Earth–Moon system can be described within the CR3BP context. Also, the neighborhood of the equilibrium points L_1 and L_2 of the Sun–Earth system has been recognized as a vantage location to place astrophysics and solar missions since the end of the '70s, with the NASA ICE/ISEE-3 mission. Since L_1 is between the Sun and the Earth, an orbit around it provides a good observation site of the Sun. Instead, L_2 is situated on the Sun–Earth line beyond the Earth and thus it is the most suitable place to put highly precise telescopes requiring great thermal stability. Also, since the Sun, the Earth and the Moon are always behind the spacecraft, orbits around L_2 ensure a constant geometry for observation with half of the entire celestial sphere available at all times. The communication system design both for L_1 and L_2 orbits is relatively simple, because they always remain close to the Earth at a distance of roughly 1.5 million km with a near-constant communication geometry. Examples of such missions are Herschel, Gaia and SOHO. Furthermore, low-energy orbital transfers especially in the Earth–Moon system and in the Jupiter's environment have been conceived[1] exploiting periodic orbits around the collinear libration points and the associated hyperbolic invariant manifolds.

Historically, the first contribution to the CR3BP is due to Euler, who introduced the synodic reference system in the XVIII century and discovered three equilibrium solutions, namely, the collinear libration points. At the same time, Lagrange determined the other two equilibria, namely, the triangular libration points. Later on, Jacobi formulated the existence of the only first integral of the problem. Other meaningful studies to the problem are due to Hill (1878), Poincaré (1899) and Birkhoff (1915).

9.2 The Three–Body Problem

The Three–Body Problem, already introduced at the beginning of Sec. 3.4.1, studies the dynamics of an isolated system composed by three punctual bodies P_i of mass m_i $(i = 1, 2, 3)$ moving due to the reciprocal gravitational attraction. According to the Law of Universal Gravitation and to the Third Law of Newton, each mass attracts the other two and is attracted by them. In an inertial reference system $\{\hat{\mathbf{i}}, \hat{\mathbf{j}}, \hat{\mathbf{k}}\}$ of origin O, the equation of motion of P_i is thus

$$m_i \frac{d^2\mathbf{r}_i}{dt^2} = G \sum_{j \neq i} \frac{m_i m_j}{r_{ij}^3} \mathbf{r}_{ij}, \tag{9.1}$$

where G is Gravitational Constant, $\mathbf{r}_i = x_i\hat{\mathbf{i}} + y_i\hat{\mathbf{j}} + z_i\hat{\mathbf{k}}$, and $\mathbf{r}_{ij}$ is the relative position of P_j with respect to P_i, namely,

$$\mathbf{r}_{ij} = \mathbf{r}_j - \mathbf{r}_i.$$

In order to solve the problem, 18 (6×3) first integrals of motion are required. In general, the n-Body Problem with $n \geq 3$ admits only 10 first integrals and thus it is not integrable. Using the same approach adopted in Secs. 1.2.1 and 1.2.3, we can prove that the angular momentum vector, namely,

$$\sum_{i=1}^{n} \mathbf{r}_i \times m_i \dot{\mathbf{r}}_i = \mathbf{h}, \tag{9.2}$$

[1].. but not realized at the moment.

is constant and that the total energy of the system, namely,

$$\frac{1}{2}\sum_{i=1}^{n} m_i v_i^2 - \frac{1}{2}G\sum_{i=1}^{n}\sum_{j=1}^{n}\frac{m_i m_j}{r_{ij}} = \mathcal{E}, \tag{9.3}$$

is conserved. Moreover, since no external forces act on the system the center of mass will move on a straight line with constant velocity. Indeed, by integrating

$$\sum_{i=1}^{n} m_i \ddot{\mathbf{r}}_i = 0, \tag{9.4}$$

we obtain

$$\sum_{i=1}^{n} m_i \dot{\mathbf{r}}_i = \mathbf{p}, \tag{9.5}$$

and, by integrating further,

$$\sum_{i=1}^{n} m_i \mathbf{r}_i = \mathbf{p}t + \mathbf{q}, \tag{9.6}$$

where $\mathbf{p}$ and $\mathbf{q}$ are two constant vectors.

We notice that, though the Three–Body Problem is not integrable, in some particular cases it is possible to compute analytical solutions, the so-called *homographic solutions*. The relative geometrical configuration among the bodies remains constant in time, or, in other words, a homographic solution can vary in time only by a scaling or a rotation. The first solutions of this kind were discovered by Euler and Lagrange and occur, respectively, when the three bodies are always situated on a straight line and remain at the vertices of an equilateral triangle. Whenever a homographic solution takes place, the three masses orbit in the same plane on a Keplerian orbit with the same focus and with the same eccentricity.

Finally, we note that if the position and velocity are specified for each body at the same given time, then the Cauchy's problem admits a unique solutions, which can be computed by means of numerical methods.

9.3 CR3BP Definitions and Equations of Motion

The CR3BP simplifies the general Three–Body Problem by assuming that one of three bodies, the third one, does not exert any force on the others (the primaries), and thus does not affect their motion. Its mass is supposed negligible. With this approximation, the study of the dynamics is reduced to the analysis of the motion of the third body, since the two primaries revolve on a Keplerian orbit around their common center of mass. In the CR3BP the Keplerian orbit is circular, which is a good approximation for most of the natural bodies in the Solar System. There also exists the *Elliptic Restricted Three–Body Problem* (ER3BP) which assumes the orbit of the primaries to be elliptic. Here we will not deal with this case, but it is important to know that the main difference

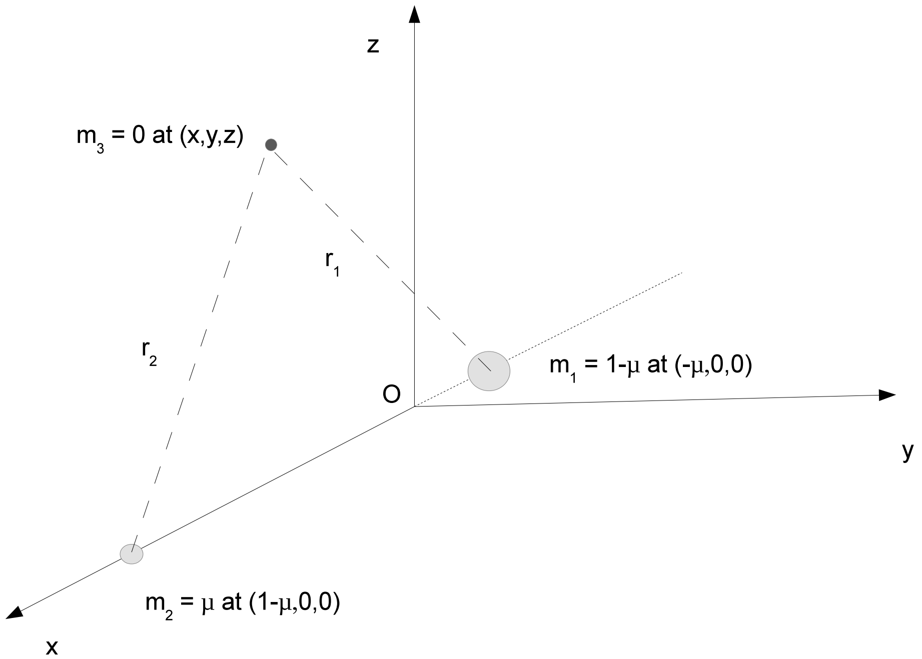

Figure 9.1: The Circular Restricted Three–Body Problem.

between the two models consists in the fact that the CR3BP is autonomous, while the ER3BP is not.

The CR3BP is usually studied in the synodic reference system, which is defined in such a way that it rotates together with the primaries around the $z-$axis with constant angular velocity w equal to the mean motion of the primaries, the origin is at the center of mass, the $x-$axis is set on the line joining the primaries, oriented in the direction of the smallest primary and the $y-$axis completes the right-handed coordinate system (see Fig. 9.1). In this way m_1 and m_2 turn out to be fixed on the $x-$axis. We notice that it is convenient to introduce this reference system in order to remove any time dependence from the equations of motion.

The relationship between the synodic reference system and the inertial one is a rotation around the z axis, namely,

$$\begin{pmatrix} x_{ine} \\ y_{ine} \\ z_{ine} \end{pmatrix} = \mathcal{R} \begin{pmatrix} x \\ y \\ z \end{pmatrix}, \tag{9.7}$$

$$\begin{pmatrix} \dot{x}_{ine} \\ \dot{y}_{ine} \\ \dot{z}_{ine} \end{pmatrix} = \mathcal{R} \begin{pmatrix} \dot{x} \\ \dot{y} \\ \dot{z} \end{pmatrix} + \dot{\mathcal{R}} \begin{pmatrix} x \\ y \\ z \end{pmatrix}, \tag{9.8}$$

being

$$\mathcal{R} = \begin{pmatrix} \cos wt & -\sin wt & 0 \\ \sin wt & \cos wt & 0 \\ 0 & 0 & 1 \end{pmatrix}. \tag{9.9}$$

It is also conventional to take advantage of non dimensional units, i.e. such that:

- $m_1 + m_2 = 1$;
- the distance between the primaries is equal to 1;
- the modulus of the angular velocity of the rotating frame is unitary.

In this way, $G = 1$ and the mass of P_1 and P_2 are, respectively,

$$\frac{m_2}{m_1 + m_2} := \mu; \qquad \frac{m_1}{m_1 + m_2} = 1 - \mu,$$

where μ is the so-called *mass parameter* and $\mu \leq 0.5$ because we assume $m_1 > m_2$. Also, it follows that the most massive body is located at $(-\mu, 0, 0)$, the second one at $(1 - \mu, 0, 0)$ (see Fig. 9.1)[2].

The transformation from non dimensional units to physical ones (subscript D) is

- length: $s_D = r_{12} s$ where r_{12} is the physical distance between the primaries;
- time: $t_D = \left(\frac{T_D}{2\pi}\right) t$ where T_D is the physical period of the relative orbit of the primaries;
- velocity: $v_D = \left(\frac{2\pi r_{12}}{T_D}\right) v$.

For the Earth–Moon system, the unit of distance equals 384400 km, the unit of time is 4.348 days and the dimensionless mass of the Moon is $\mu = \frac{m_2}{m_1+m_2} = 1.2150582 \times 10^{-2}$. For the Sun–Earth+Moon system 1 AU, 58.131 days and $\mu = 3.0404234 \times 10^{-6}$.

[2] Actually, some authors use a different convention, that is, m_1 at μ and m_2 at $-1 + \mu$.

The motion of the massless particle P is driven by the differential equation[3]

$$\frac{\partial^2 \mathbf{r}}{\partial t^2} + \dot{\mathbf{w}} \times \mathbf{r} + 2\mathbf{w} \times \frac{\partial \mathbf{r}}{\partial t} + \mathbf{w} \times (\mathbf{w} \times \mathbf{r}) = \frac{\partial \mathcal{U}}{\partial \mathbf{r}}, \tag{9.12}$$

where $\mathbf{r} = (x, y, z)$ is the radius vector associated with P, $\mathcal{U}$ the gravitational potential

$$\mathcal{U} = \frac{1-\mu}{r_1} + \frac{\mu}{r_2}, \tag{9.13}$$

and r_1 and r_2 represent the distance to P from P_1 and P_2, respectively,

$$\begin{aligned} r_1 &= [(x+\mu)^2 + y^2 + z^2]^{\frac{1}{2}}, \\ r_2 &= [(x-1+\mu)^2 + y^2 + z^2]^{\frac{1}{2}}. \end{aligned}$$

Since $\mathbf{w}$ is constant, we get to

$$\frac{\partial^2 \mathbf{r}}{\partial t^2} + 2\mathbf{w} \times \frac{\partial \mathbf{r}}{\partial t} + \mathbf{w} \times (\mathbf{w} \times \mathbf{r}) = \frac{\partial \mathcal{U}}{\partial \mathbf{r}}. \tag{9.14}$$

We notice that the second and third term on the left hand side of (9.14) represent the accelerations due to the Coriolis and the centrifugal forces.

In the three coordinates, the equations of motion are

$$\begin{aligned} \ddot{x} - 2\dot{y} &= \frac{\partial \mathcal{U}^*}{\partial x} = x - \frac{(1-\mu)}{r_1^3}(x+\mu) - \frac{\mu}{r_2^3}(x-1+\mu), \\ \ddot{y} + 2\dot{x} &= \frac{\partial \mathcal{U}^*}{\partial y} = y - \frac{(1-\mu)}{r_1^3} y - \frac{\mu}{r_2^3} y, \\ \ddot{z} &= \frac{\partial U^*}{\partial z} = -\frac{(1-\mu)}{r_1^3} z - \frac{\mu}{r_2^3} z, \end{aligned} \tag{9.15}$$

where $\mathcal{U}^*$ is the effective potential defined as

$$\mathcal{U}^* = \mathcal{U} + \mathcal{U}_c, \tag{9.16}$$

[3] Recall that in a rotating reference system, the following expressions hold:

$$\frac{d}{dt} = \frac{\partial}{\partial t} + \mathbf{w}\times, \tag{9.10}$$

$$\begin{aligned} \frac{d^2\mathbf{r}}{dt^2} &= \left(\frac{\partial}{\partial t} + \mathbf{w}\times\right)\left(\frac{\partial}{\partial t} + \mathbf{w}\times\right)\mathbf{r} = \\ &= \left(\frac{\partial}{\partial t} + \mathbf{w}\times\right)\left(\frac{\partial \mathbf{r}}{\partial t} + \mathbf{w}\times\mathbf{r}\right) = \\ &= \frac{\partial}{\partial t}\left(\frac{\partial \mathbf{r}}{\partial t} + \mathbf{w}\times\mathbf{r}\right) + \mathbf{w}\times\left(\frac{\partial \mathbf{r}}{\partial t} + \mathbf{w}\times\mathbf{r}\right) = \\ &= \frac{\partial^2 \mathbf{r}}{\partial t^2} + \frac{\partial \mathbf{w}}{\partial t}\times\mathbf{r} + \mathbf{w}\times\frac{\partial \mathbf{r}}{\partial t} + \mathbf{w}\times\frac{\partial \mathbf{r}}{\partial t} + \mathbf{w}\times(\mathbf{w}\times\mathbf{r}) = \\ &= \frac{\partial^2 \mathbf{r}}{\partial t^2} + \frac{\partial \mathbf{w}}{\partial t}\times\mathbf{r} + 2\mathbf{w}\times\frac{\partial \mathbf{r}}{\partial t} + \mathbf{w}\times(\mathbf{w}\times\mathbf{r}) = \\ &= \frac{\partial^2 \mathbf{r}}{\partial t^2} + \dot{\mathbf{w}}\times\mathbf{r} + 2\mathbf{w}\times\frac{\partial \mathbf{r}}{\partial t} + \mathbf{w}\times(\mathbf{w}\times\mathbf{r}). \end{aligned} \tag{9.11}$$

being $\mathcal{U}_c$ the centrifugal potential

$$\mathcal{U}_c = \frac{1}{2}(\mathbf{w}\times\mathbf{r})\cdot(\mathbf{w}\times\mathbf{r}) = \frac{1}{2}(x^2+y^2). \tag{9.17}$$

It is interesting to note the existence of some important symmetries:

- with respect to the $\{x-y\}$ plane, if $(x(t), y(t), z(t))$ is a solution of (9.15), then $(x(t), y(t), -z(t))$ also is;
- with respect to time, if $(x(t), y(t), z(t))$ is a solution of (9.15), then $(x(-t), -y(-t), z(-t))$ also is;
- with respect to the mass parameter, if $(x(t), y(t), z(t))$ is a solution of (9.15) for μ, then $(-x(t), -y(t), z(t))$ is a solution for $1-\mu$.

9.4 The Jacobi Integral and the Equilibrium Points

The CR3BP admits one first integral of motion, the *Jacobi integral*, which is defined as

$$C = 2\mathcal{U}^* - ||\dot{\mathbf{r}}||^2 = (x^2+y^2) + \frac{2(1-\mu)}{r_1} + \frac{2\mu}{r_2} - \left(\dot{x}^2+\dot{y}^2+\dot{z}^2\right). \tag{9.18}$$

Indeed, from (9.15) we have

$$\ddot{x}\dot{x} + \ddot{y}\dot{y} + \ddot{z}\dot{z} = \frac{\partial\mathcal{U}^*}{\partial x}\dot{x} + \frac{\partial\mathcal{U}^*}{\partial y}\dot{y} + \frac{\partial\mathcal{U}^*}{\partial z}\dot{z} = \frac{\partial\mathcal{U}^*}{\partial t},$$

which, by integration, gives

$$\dot{x}^2 + \dot{y}^2 + \dot{z}^2 = 2\mathcal{U}^* - C.$$

The Jacobi integral can be interpreted as the energy of the particle in the rotating frame. It can also be proved that

$$C = -2\mathcal{H}, \tag{9.19}$$

where $\mathcal{H}$ is the Hamiltonian function associated with the problem.
Considering that $||\dot{\mathbf{r}}||^2 \geq 0$, (9.18) states that

$$2\mathcal{U}^* \geq C.$$

In the particular case where the kinetic energy of the particle is zero, the function

$$f(\mathbf{r}, C) = 2\mathcal{U}^* - C = 0 \tag{9.20}$$

identifies, as a function of the value of C and thus of the initial conditions, the boundaries in the phase space within which the particle can move. In other words, (9.20) defines the zero velocity curves, beyond them the kinetic energy turns negative and the motion is not allowed. The regions where the particle can orbit are known are *Hill's regions* (see

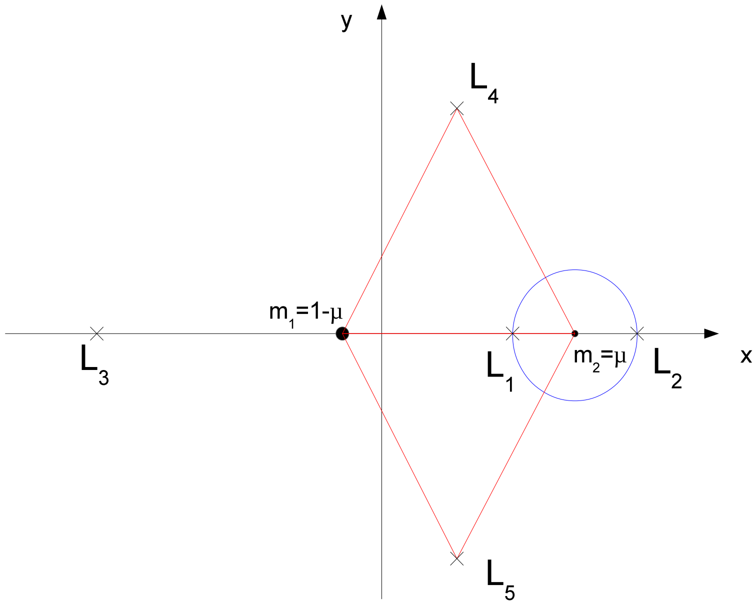

Figure 9.2: The equilibrium points of the Circular Restricted Three–Body Problem and the Hill's sphere (blue).

Fig. 9.3). From (9.19) it can be derived that high values of C correspond to low values of energy and the motion is bounded in the neighborhood of the primaries. Clearly, the more the energy increases the more the forbidden regions shrink and the particle is free to recede from P_1 and P_2.

In the synodic reference system there exist five equilibrium points (see Fig. 9.2), the so-called *Lagrange* or *libration points* L_i $(i = 1, 5)$, which can be computed imposing velocity $\{\dot{x}, \dot{y}, \dot{z}\}$ and acceleration $\{\ddot{x}, \ddot{y}, \ddot{z}\}$ to be null in the equations of motion (9.15). This is,

$$
\begin{aligned}
0 &= x - \frac{(1-\mu)}{r_1^3}(x+\mu) - \frac{\mu}{r_2^3}(x-1+\mu), \\
0 &= \left[1 - \frac{(1-\mu)}{r_1^3} - \frac{\mu}{r_2^3}\right] y, \\
0 &= \left[-\frac{(1-\mu)}{r_1^3} - \frac{\mu}{r_2^3}\right] z.
\end{aligned}
\tag{9.21}
$$

First, we notice that the third equation always implies that $z = 0$[4], that is, the equilibrium points always lie in the $\{x - y\}$ plane. The second equation, instead, is verified in two cases:

[4]The other term on the right-hand side of the third equation cannot be zero, because r_1 and r_2 are positive quantities.

1. if $y = 0$;
2. if $y \neq 0$ and $\left[1 - \frac{(1-\mu)}{r_1^3} - \frac{\mu}{r_2^3}\right] = 0$.

The first condition corresponds to the *collinear equilibrium points* L_1, L_2 and L_3, the second to the *triangular equilibrium points* L_4 and L_5. L_1, L_2 and L_3 are thus said collinear points because they lie on the $x-$axis. Setting $y = 0$, $z = 0$ the equation of motion corresponding to x becomes

$$x - \frac{(1-\mu)(x+\mu)}{|x+\mu|^3} - \frac{\mu(x-1+\mu)}{|x-1+\mu|^3} = 0. \tag{9.22}$$

Let us introduce the variable u as

$$u = x - 1 + \mu. \tag{9.23}$$

We have

$$x + \mu = u + 1, \tag{9.24}$$

and therefore

$$\begin{aligned} r_1 &= |x+\mu| = |u+1|, \\ r_2 &= |x+\mu-1| = |u|. \end{aligned}$$

With respect to the variable u, (9.22) reads

$$u + 1 - \mu - \frac{(1-\mu)(u+1)}{|u+1|^3} - \frac{\mu u}{|u|^3} = 0,$$

that is,

$$u + 1 - \mu - \frac{(1-\mu)s_1}{(u+1)^2} - \frac{\mu s_0}{u^2} = 0, \tag{9.25}$$

where $s_0 = sgn(u)$ and $s_1 = sgn(u+1)$.
After some manipulation, (9.25) can be put into an algebraic equation of degree 5 in u, namely the *Lagrange quintic equation*,

$$u^2(1 - s_1 + 3u + 3u^2 + u^3) = \mu[s_0 + 2s_0 u + (1 + s_0 - s_1)u^2 + 2u^3 + u^4]. \tag{9.26}$$

We can distinguish among different combinations of s_0 and s_1 as a function of the position of the collinear point, by identifying three intervals on the $x-$axis, namely,

$$\begin{array}{lclcl} L_3 & : & (s_0, s_1) = (-1,-1) & \text{if} & x < -\mu, \\ L_1 & : & (s_0, s_1) = (-1,1) & \text{if} & -\mu < x < 1-\mu, \\ L_2 & : & (s_0, s_1) = (1,1) & \text{if} & x > 1-\mu. \end{array}$$

Depending on the interval Eq. (9.26) becomes, respectively,

$$\begin{array}{lcl} L_3 & : & u^2(u^3 + 3u^2 + 3u + 2) = \mu(u^4 + 2u^3 + u^2 - 2u - 1), \\ L_1 & : & u^2(u^3 + 3u^2 + 3u) = \mu(u^4 + 2u^3 - u^2 - 2u - 1), \\ L_2 & : & u^2(u^3 + 3u^2 + 3u) = \mu(u^4 + 2u^3 + u^2 + 2u + 1). \end{array} \tag{9.27}$$

Each of these three equations admits one real solution and two pair of conjugate complex solutions and thus we have three collinear equilibrium points. It is possible to find an analytical solution for (9.27), but only in terms of elliptic functions. By means of numerical algorithms, we can instead compute valid approximate solutions, starting from the following initial guess for u:

$$\begin{aligned} L_1 &: \quad u_0 = \alpha\Big(1 - \frac{\alpha}{3} - \frac{\alpha^2}{9}\Big), \\ L_2 &: \quad u_0 = \alpha\Big(1 + \frac{\alpha}{3} - \frac{\alpha^2}{9}\Big), \\ L_3 &: \quad u_0 = \beta - 2, \end{aligned} \tag{9.28}$$

where

$$\alpha = \left[\frac{\mu}{3(1-\mu)}\right]^{\frac{1}{3}} \approx \left[\frac{\mu}{3}\right]^{\frac{1}{3}}, \qquad \beta = \nu\Big(1 + \frac{23}{84}\nu^2\Big), \qquad \nu = \left(\frac{7}{12}\right)\mu.$$

In the CR3BP framework, the gravitational sphere is defined as the sphere having radius $r_H = \alpha$. It is usually called *Hill's sphere* and its boundaries are the boundaries between the realm of m_1 and the one of m_2 (see Fig. 9.2). As a side note, one may wonder when to adopt the CR3BP is preferable to apply the third-body perturbation (see Sec. 3.4) to the Two–Body Problem. The exact point of transition between the two approaches is not well–defined actually, but we can say that the latter represents a valid approximation as long as the distance between the probe and a given massive body is much less than the distance between the two primaries (see Eq. (3.82)).

The remaining two equilibrium points, L_4 and L_5, also belong to the xy plane and are obtained by solving

$$\begin{aligned} 0 &= x - \frac{(1-\mu)(x+\mu)}{r_1^3} - \frac{\mu(x-1+\mu)}{r_2^3}, \\ 0 &= 1 - \frac{(1-\mu)}{r_1^3} - \frac{\mu}{r_2^3}. \end{aligned}$$

Let us multiply the second equation by $x - 1 + \mu$ and subtract it from the first one, namely,

$$\cancel{x} - \frac{(1-\mu)(x+\mu)}{r_1^3} - \cancel{\frac{\mu(x-1+\mu)}{r_2^3}} - \cancel{x} + 1 - \mu + \frac{(1-\mu)(x-1+\mu)}{r_1^3} + \cancel{\frac{\mu(x-1+\mu)}{r_2^3}} = 0,$$

this is,

$$\frac{1-\mu}{r_1^3} = 1 - \mu,$$

which gives $r_1 = 1$.

Let us multiply the second equation by $x+\mu$ and subtract it from the first one, namely,

$$\cancel{x} - \cancel{\frac{(1-\mu)(x+\mu)}{r_1^3}} - \frac{\mu(x-1+\mu)}{r_2^3} - \cancel{x} - \mu + \cancel{\frac{(1-\mu)(x+\mu)}{r_1^3}} + \frac{\mu(x+\mu)}{r_2^3} = 0,$$

this is,

$$\frac{\mu}{r_2^3} \quad = \quad \mu,$$

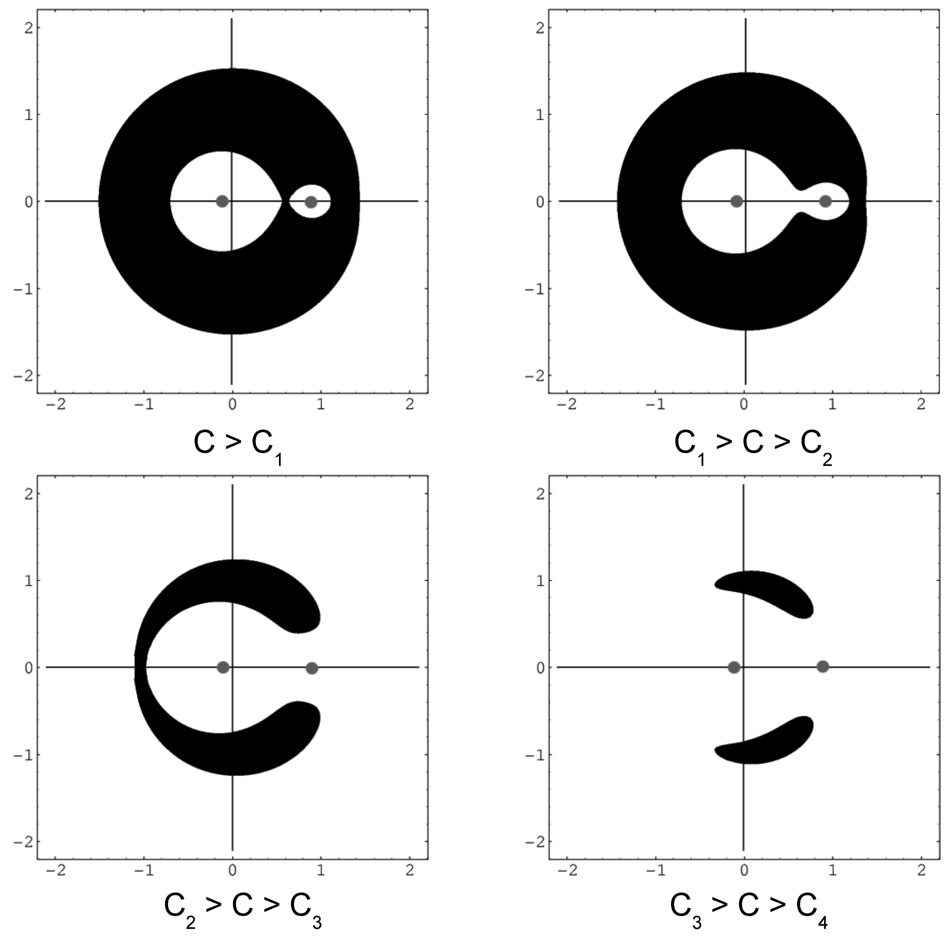

Figure 9.3: According to the value of the Jacobi constant, the particle is not allowed to move within certain regions, the black ones. The grey points represent the primaries, the white regions are the Hill's regions. If $C < C_5 = C_4$ the motion can take place everywhere.

which gives $r_2 = 1$. The triangular points are thus located at the vertices of two equilateral triangles of unitary side, namely,

$$
\begin{aligned}
L_4 \quad &: \qquad \left(\frac{1}{2} - \mu, \frac{\sqrt{3}}{2}, 0\right), \\
L_5 \quad &: \qquad \left(\frac{1}{2} - \mu, -\frac{\sqrt{3}}{2}, 0\right),
\end{aligned}
$$

being the primaries located on the other two vertices (see Fig. 9.3).
Finally, if C_i denotes the Jacobi constant corresponding to the L_i point, then

$$C_5 = C_4 < C_3 < C_2 < C_1,$$

which tells us that L_1 is the point of minimum energy. As mentioned before, for a given value of the mass parameter μ, depending on the value of C the Hill's regions correspond to five possible configurations (see Fig. 9.3).

9.5 The Stability of the Libration Points

The collinear libration points have a *centre* $\times$ *centre* $\times$ *saddle* behavior for any value of μ and are thus unstable, while the triangular points are characterized by a *centre* $\times$ *centre* $\times$ *centre* behavior for $\mu < \mu_R$, being μ_R the *Routh's mass parameter*, whose value will be specified later. To prove that, we perform a linearization around a given equilibrium point L_i, that is, we assume that the particle is displaced with respect to L_i by a small amount and we study its motion. Let

$$\mathbf{X}' = (x_{L_i} + x', y_{L_i} + y', z_{L_i} + z', \dot{x}_{L_i} + \dot{x}', \dot{y}_{L_i} + \dot{y}', \dot{z}_{L_i} + \dot{z}'),$$

the state vector of the massless body at some value of time. The equations of motion (9.15) applied to $\mathbf{X}'$ read

$$
\begin{aligned}
\ddot{x}_{L_i} - 2\dot{y}_{L_i} + \ddot{x}' - 2\dot{y}' &= \frac{\partial \mathcal{U}^*}{\partial x}(\mathbf{X}') = \frac{\partial \mathcal{U}^*}{\partial x}(L_i) + x'\frac{\partial^2 \mathcal{U}^*}{\partial x \partial x}(L_i) + y'\frac{\partial^2 \mathcal{U}^*}{\partial x \partial y}(L_i) + z'\frac{\partial^2 \mathcal{U}^*}{\partial x \partial z}(L_i), \\
\ddot{y}_{L_i} + 2\dot{x}_{L_i} + \ddot{y}' + 2\dot{x}' &= \frac{\partial \mathcal{U}^*}{\partial y}(\mathbf{X}') = \frac{\partial \mathcal{U}^*}{\partial y}(L_i) + x'\frac{\partial^2 \mathcal{U}^*}{\partial y \partial x}(L_i) + y'\frac{\partial^2 \mathcal{U}^*}{\partial y \partial y}(L_i) + z'\frac{\partial^2 \mathcal{U}^*}{\partial y \partial z}(L_i), \\
\ddot{z}_{L_i} + \ddot{z}' &= \frac{\partial U^*}{\partial z}(\mathbf{X}') = \frac{\partial \mathcal{U}^*}{\partial z}(L_i) + x'\frac{\partial^2 \mathcal{U}^*}{\partial z \partial x}(L_i) + y'\frac{\partial^2 \mathcal{U}^*}{\partial z \partial y}(L_i) + z'\frac{\partial^2 \mathcal{U}^*}{\partial z \partial z}(L_i),
\end{aligned}
$$

where we have applied a Taylor expansion at L_i with respect to the position and the derivatives appearing in the system are

$$
\begin{aligned}
\mathcal{U}^*_{xx} \equiv \frac{\partial^2 \mathcal{U}^*}{\partial x \partial x} &= 1 - \frac{1-\mu}{r_1^3} - \frac{\mu}{r_2^3} + 3(1-\mu)\frac{(x+\mu)^2}{r_1^5} + 3\mu\frac{(x-1+\mu)^2}{r_2^5}, \\
\mathcal{U}^*_{yy} \equiv \frac{\partial^2 \mathcal{U}^*}{\partial y \partial y} &= 1 - \frac{1-\mu}{r_1^3} - \frac{\mu}{r_2^3} + 3(1-\mu)\frac{y^2}{r_1^5} + 3\mu\frac{y^2}{r_2^5}, \\
\mathcal{U}^*_{zz} \equiv \frac{\partial^2 \mathcal{U}^*}{\partial z \partial z} &= -\frac{1-\mu}{r_1^3} - \frac{\mu}{r_2^3} + 3(1-\mu)\frac{z^2}{r_1^5} + 3\mu\frac{z^2}{r_2^5}, \\
\mathcal{U}^*_{xy} \equiv \frac{\partial^2 \mathcal{U}^*}{\partial x \partial y} = \frac{\partial^2 \mathcal{U}^*}{\partial y \partial x} &= 3\left[(1-\mu)\frac{x+\mu}{r_1^5} + \mu\frac{x-1+\mu}{r_2^5}\right] y, \\
\mathcal{U}^*_{xz} \equiv \frac{\partial^2 \mathcal{U}^*}{\partial x \partial z} = \frac{\partial^2 \mathcal{U}^*}{\partial z \partial x} &= 3\left[(1-\mu)\frac{x+\mu}{r_1^5} + \mu\frac{x-1+\mu}{r_2^5}\right] z, \\
\mathcal{U}^*_{yz} \equiv \frac{\partial^2 \mathcal{U}^*}{\partial y \partial z} = \frac{\partial^2 \mathcal{U}^*}{\partial z \partial y} &= 3\left[\frac{1-\mu}{r_1^5} + \frac{\mu}{r_2^5}\right] yz.
\end{aligned}
\tag{9.29}
$$

We notice that for the last three expressions we exploit the symmetry characterizing the equations of motion ($\mathcal{U}^*_{xy} = \mathcal{U}^*_{yx}$ etc.) and that those three derivatives are zero for the collinear equilibrium points, while for the triangular points only the last two cancel out. Moreover, we know that for any equilibrium point we have

$$
z_{L_i} = \dot{x}_{L_i} = \dot{y}_{L_i} = \dot{z}_{L_i} = \frac{\partial \mathcal{U}^*}{\partial x}(L_i) = \frac{\partial \mathcal{U}^*}{\partial y}(L_i) = \frac{\partial \mathcal{U}^*}{\partial z}(L_i) = 0,
$$

and thus the linearized equations are

$$
\begin{aligned}
\ddot{x}' - 2\dot{y}' &= x'\mathcal{U}^*_{xx}(L_i) + y'\mathcal{U}^*_{xy}(L_i), \\
\ddot{y}' + 2\dot{x}' &= x'\mathcal{U}^*_{xy}(L_i) + y'\mathcal{U}^*_{yy}(L_i), \\
\ddot{z}' &= z'\mathcal{U}^*_{zz}(L_i).
\end{aligned}
\tag{9.30}
$$

The general solution of this system is given by a linear combination of exponential functions of the type $e^{\lambda t}$. If any of the λ coefficients is real, then the equilibrium point will be unstable, that is, any small displacement from L_i will bring the particle to depart exponentially from L_i; otherwise, if all the coefficients are purely imaginary, then the equilibrium point will be stable and the particle will oscillate around it on a bounded orbit.

9.5.1 Collinear Points

Let us consider the case of the collinear libration points, for which we have

$$
\begin{aligned}
\ddot{x}' - 2\dot{y}' &= x'\mathcal{U}^*_{xx}(L_i), \\
\ddot{y}' + 2\dot{x}' &= y'\mathcal{U}^*_{yy}(L_i), && (9.31) \\
\ddot{z}' &= z'\mathcal{U}^*_{zz}(L_i), && (9.32)
\end{aligned}
$$

with

$$
\begin{aligned}
\mathcal{U}^*_{xx}(L_i) &= 1+2\frac{1-\mu}{r_1^3}+2\frac{\mu}{r_2^3}, \\
\mathcal{U}^*_{yy}(L_i) &= 1-\frac{1-\mu}{r_1^3}-\frac{\mu}{r_2^3}, \\
\mathcal{U}^*_{zz}(L_i) &= -\frac{1-\mu}{r_1^3}-\frac{\mu}{r_2^3},
\end{aligned}
\tag{9.33}
$$

because $r_1 = x+\mu$ and $r_2 = x-1+\mu$ at these points.

First, we notice that the equation in z is uncoupled from the other two, because there do not appear x or y velocity components. Therefore, the motion on the $z-$direction will be a harmonic motion. For the other two components, we look for a solution of the type

$$x' = A_1 e^{\lambda t}, \qquad y' = A_2 e^{\lambda t},$$

which, once plugged into (9.31) and noting that $e^{\lambda t} > 0$, gives

$$
\begin{aligned}
\left[\lambda^2 - \mathcal{U}^*_{xx}(L_i)\right] A_1 - 2\lambda A_2 &= 0, \\
2\lambda A_1 + \left[\lambda^2 - \mathcal{U}^*_{yy}(L_i)\right] A_2 &= 0.
\end{aligned}
$$

The characteristic equation of this system is

$$\lambda^4 + 2C\lambda^2 - D = 0, \tag{9.34}$$

where

$$
\begin{aligned}
C &= 2-\frac{1}{2}\left[\mathcal{U}^*_{xx}(L_i)+\mathcal{U}^*_{yy}(L_i)\right], \\
D &= -\mathcal{U}^*_{xx}(L_i)\mathcal{U}^*_{yy}(L_i).
\end{aligned}
$$

The solutions of (9.34) are

$$\lambda_{1,2,3,4} = \pm\sqrt{-C\pm\sqrt{C^2+D}}, \tag{9.35}$$

which consists of two real and two purely imaginary numbers, because $D > 0$. Indeed, from (9.33) $\mathcal{U}^*_{xx}(L_i) > 0$ and $\mathcal{U}^*_{yy}(L_i) < 0$, because

$$
\begin{aligned}
\mathcal{U}^*_{yy}(L_i) &= 1-\frac{1-\mu}{r_1^3}-\frac{\mu}{r_2^3} \\
&= 1-\frac{1}{r_1}\left(\frac{1}{r_1^2}-\frac{\mu}{r_1^2}\right)-\frac{\mu}{r_2^3},
\end{aligned}
$$

and the first equation in (9.21) evaluated at any collinear point gives

$$\frac{1}{r_1^2}-\frac{\mu}{r_1^2} = r_1-\mu-\frac{\mu}{r_2^2},$$

because $r_1 = x + \mu$ and $r_2 = x - 1 + \mu$ at these points. So, we obtain

$$\begin{aligned}
\mathcal{U}^*_{yy}(L_i) &= 1 - \frac{1}{r_1}\left(r_1 - \mu - \frac{\mu}{r_2^2}\right) - \frac{\mu}{r_2^3} \\
&= \frac{\mu}{r_1}\left(1 + \frac{1}{r_2^2} - \frac{r_1}{r_2^3}\right) \\
&= \frac{\mu}{r_1}\left(1 + \frac{1}{r_2^2} - \frac{1 + r_2}{r_2^3}\right) \\
&= \frac{\mu}{r_1}\left(1 - \frac{1}{r_2^3}\right),
\end{aligned}$$

which is negative because $r_2 < 1$, as it can be proved by using the first approximations for L_i $(i = 1, 2, 3)$ given in the previous section in terms of α, β and ν.

By combining the solutions (9.35) with the harmonic motion in the $z-$component, we prove that the linear behavior associated with the collinear libration points is of the type *centre* $\times$ *centre* $\times$ *saddle*. As introduced above, the central components determine the existence of bounded orbits (central orbits) in the neighborhood of the collinear libration points: they can be periodic or quasi-periodic, i.e. such that they occur on an invariant torus. There exist different types of central orbits, the most exploited ones being planar and vertical Lyapunov periodic orbits, halo periodic orbits, Lissajous quasi-periodic orbits and quasi-halo quasi-periodic orbits around L_1 and L_2 (see Fig. 9.4). On the other hand, the saddle component gives rise to the so-called stable and unstable invariant manifolds, or hyperbolic invariant manifolds associated with a nominal central orbit. They look like tubes of asymptotic trajectories tending to, or departing from the corresponding orbit, respectively. Each hyperbolic manifold has two branches, one going towards μ, the other towards $1 - \mu$ (see Fig. 9.5). For practical applications, at the moment central orbits in the Sun–Earth system are exploited as nominal orbits for astrophysics and solar missions, they require a periodic station-keeping procedure because of their characteristic instability. The trajectories on the corresponding stable invariant manifold are usually considered to reach such operational orbits at almost zero-cost from the Earth.

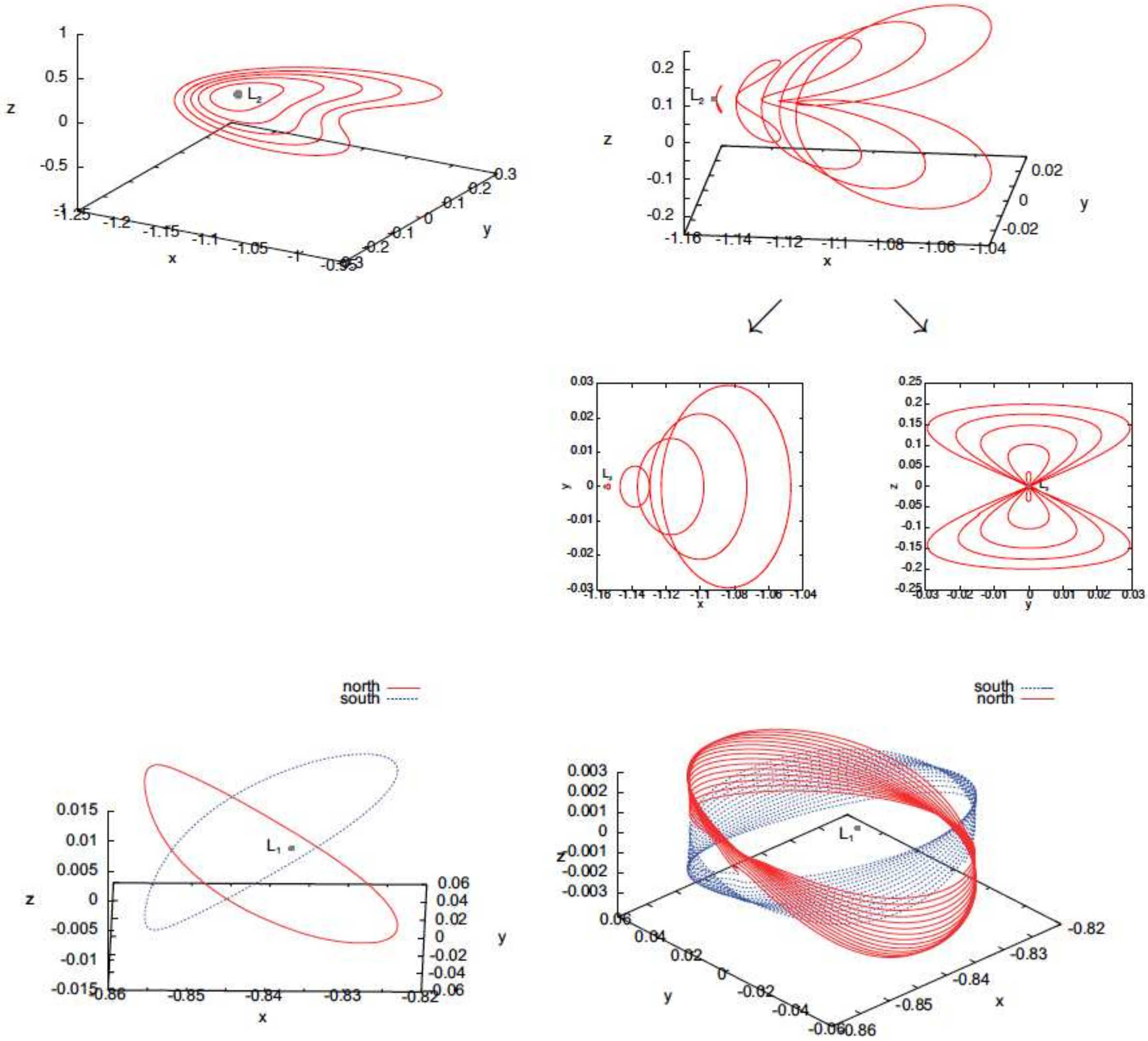

Figure 9.4: Examples of central orbits in the neighborhood of L_2 in the Earth–Moon system. Top: planar (left) and vertical (right) Lyapunov orbits. Bottom: halo (left) and quasi-halo orbits. Each orbit on the top plots corresponds to a different value of C. For these figures m_1 at μ and m_2 at $-1+\mu$.

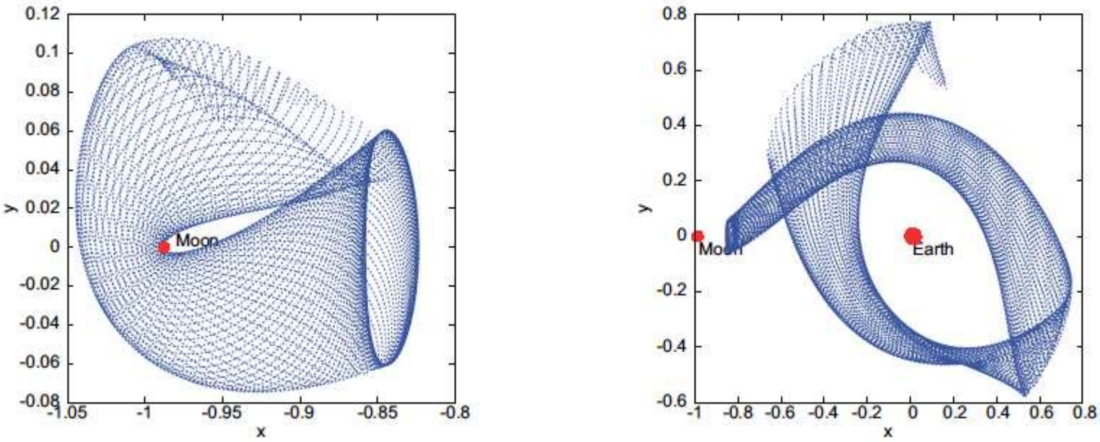

Figure 9.5: Example of stable invariant manifold of a nominal libration point orbit in the Earth–Moon system. The two figures represent the two branches of the manifold. For these figures m_1 at μ and m_2 at $-1+\mu$.

9.5.2 Triangular Points

For the triangular libration points, the linearized system reads

$$\begin{aligned} \ddot{x}' - 2\dot{y}' &= \frac{3}{4}x' + \frac{\sqrt{27}}{4}(1-2\mu)y', \\ \ddot{y}' + 2\dot{x}' &= \frac{\sqrt{27}}{4}(1-2\mu)x' + \frac{9}{4}y', \qquad (9.36) \\ \ddot{z}' &= -z'. \qquad (9.37) \end{aligned}$$

On the $z-$direction, we have also in this case a harmonic oscillation, that is, any equilibrium point is stable with respect to the $z-$component. For the other two components, the characteristic equation is now

$$\lambda^4 + \lambda^2 + \frac{27}{4}\mu(1-\mu) = 0, \tag{9.38}$$

whose roots are

$$\lambda_{1,2,3,4} = \pm\sqrt{\frac{-1 \pm \sqrt{1-27\mu(1-\mu)}}{2}}. \tag{9.39}$$

If

$$1 - 27\mu(1-\mu) \geq 0,$$

then any λ will a purely imaginary number and thus the point will be stable, with a linear behavior of the type *centre* $\times$ *centre* $\times$ *centre*. Such condition is verified if

$$\mu \geq \frac{1}{2} + \sqrt{\frac{23}{108}},$$

or

$$\mu \leq \frac{1}{2} - \sqrt{\frac{23}{108}}.$$

Since we assumed $\mu \leq 0.5$, then only the second condition will be acceptable. It defines the so-called *Routh's mass parameter*, namely,

$$\mu_R = \frac{1}{2} - \sqrt{\frac{23}{108}} \approx 0.0385.$$

In the Solar System, $\mu < \mu_R$ for all the Sun-planet systems.